

A Practical Sharpe-Ratio Diagnostic for Portfolio Backtests

Abstract

Backtests often report strategy Sharpe ratios from panels of selected asset returns. In practice, however, a portfolio earns one aggregate return per period. Treating many same-date selected rows as independent performance observations can therefore make a backtest look more precise than the investor-level return series supports. This article presents a practical Sharpe-ratio diagnostic for portfolio backtests. The procedure aggregates selected rows into date-level portfolio returns, tests Sharpe evidence on that date series using robust inference, and reports how much apparent precision came from row-level counting. Public Kenneth French and AQR examples show how the check preserves strong benchmark momentum evidence while flagging weaker panel claims. Simulation stress tests show that row-naïve Sharpe tests can reject about 35 percent of the time at a nominal 5 percent level under dependent null designs with no true edge. The contribution is a backtest-review tool for portfolio researchers, asset managers, and investment due-diligence teams, not a new trading rule or asset-pricing anomaly.

Keywords: Sharpe ratio; backtesting; performance measurement; portfolio evaluation; panel data; false precision

1 Introduction: a practical check for false precision

Systematic-investment research often starts with a panel. A researcher scores many assets or portfolios on each date, selects the rows that pass a signal screen, and reports a Sharpe ratio for the resulting rule. This workflow is natural in stock selection, factor screening, anomaly testing, and automated strategy development. The practical reporting problem is that the selected rows are not the payoff boundary. A portfolio return is realized once per date.

If selected asset-date rows are treated as independent observations, a backtest can look much more precise than the realized date-level portfolio evidence supports. Fifty selected rows on the same date are not fifty independent performance observations when

those rows share market shocks, factor shocks, or common implementation conditions. The number that matters for Sharpe-ratio inference is the date-return series generated by the declared portfolio rule.

This article is intended as a practical reporting tool for portfolio researchers and asset managers who evaluate panel-based trading signals. It is not a new trading rule, not a new anomaly, and not a claim that one new statistic should replace standard robust inference. The contribution is a reproducible backtest-review workflow for checking whether Sharpe-ratio evidence is reported on the return series an investor could actually receive.

The workflow asks three questions. First, does the signal survive after selected asset-date rows are aggregated into one active portfolio return per date? Second, does the date-level Sharpe remain significant under robust inference? Third, how much precision was created by the row-pooled analysis relative to the date-level portfolio evidence? The answer to the third question is reported as a Sharpe uncertainty- inflation check. It is a descriptive warning label, not the rejection rule.

The public examples use Kenneth French and AQR data because they are transparent and reproducible. The examples are deliberately modest: canonical French winner-minus-loser (WML) momentum survives the diagnostic, a weaker French Size/BM skipped-momentum stress case fails, and AQR factor series are treated as time-series benchmarks because public constituent-level rows are not available. The simulation evidence then asks how large the false-precision problem can be in a controlled setting. Under dependent null designs with no true date-level edge, row-naive tests reject about 35 percent of the time at nominal 5 percent size, while date-level HAC and dependent-bootstrap procedures remain much closer to nominal size.

The intended reader is a portfolio evaluator who must decide whether a reported backtest is strong enough to justify further research, paper trading, or production due diligence. For that reader, the central question is not whether every possible dependence model has been solved. It is whether the reported Sharpe evidence was attached to the payoff series that an investor could actually receive. A diagnostic that prevents row counts from being mistaken for independent performance dates can improve the review process even when the final investment decision still depends on costs, liquidity, capacity, and business constraints.

The paper is organized around use rather than theory. Section 2 explains the date-level payoff boundary. Section 3 gives the diagnostic checklist for portfolio evaluators. Section 4 reports worked public examples. Section 5 gives the simulation stress test. Section 6 summarizes practical reporting guidance and limitations. Algebra and extended tables are left to the technical appendix.

2 The practical problem: rows are not the payoff boundary

A panel backtest usually begins with observations $(t, i, s_{t,i}, r_{t+1,i})$, where t indexes dates, i indexes assets or portfolios, $s_{t,i}$ is a signal, and $r_{t+1,i}$ is the next-period return. A rule selects a set S_t of rows on each date and assigns portfolio weights $w_{t,i}$. The economic performance object is the date-level portfolio return

$$R_{t+1} = \sum_{i \in S_t} w_{t,i} r_{t+1,i}.$$

In words, the selected rows are inputs to one portfolio return for that date. The date-return series $\{R_t\}$, not the multiplied set of selected rows, is the object on which Sharpe-ratio evidence should be tested.

The row-naïve alternative instead treats selected $r_{t+1,i}$ as if they were independent performance observations. That can be a useful diagnostic comparator because it shows what a common but unsafe workflow would conclude. It should not be the confirmatory test when the investment claim is about a portfolio return.

The issue is a portfolio-performance version of a familiar design-effect problem. When observations within a cluster share common shocks, the effective amount of information is smaller than the raw row count. In portfolio panels, the natural cluster is often the decision date. The same-date selected rows may diversify idiosyncratic noise, but they do not create additional time-series observations for Sharpe inference. This is why a rule can have thousands of selected rows and still only a few thousand, or a few hundred, economically relevant return dates.

The Sharpe uncertainty-inflation check summarizes the reporting gap:

$$\text{Sharpe uncertainty inflation} = \frac{\text{date-level long-run variance of the Sharpe estimate}}{\text{row-pooled IID variance of the Sharpe estimate}}.$$

In words, the check compares the uncertainty attached to the realized date-level portfolio return series with the uncertainty a row-pooled independent-observation report would attach to the selected rows. A value above one indicates that the row-pooled report is too precise relative to the date-level portfolio evidence. The diagnostic is descriptive; the reported decision still comes from robust date-level inference, menu adjustment, placebo checks, and turnover-cost sensitivity. The technical appendix gives the formal version for readers who want the algebra.

3 Diagnostic workflow for portfolio evaluators

For a portfolio researcher reviewing a panel backtest, the diagnostic can be implemented as the following checklist.

First, define the economic portfolio return for each date. Selected asset-date rows should be converted into one active portfolio return per date using the declared weighting rule. If the rule is long- short, the gross-exposure convention should be stated before results are examined.

Second, compute Sharpe evidence on that date-return series. The date-level return is the payoff boundary. The Sharpe ratio should be the annualized mean of that return series divided by its period volatility, using a source-appropriate annualization factor.

Third, report row-naive results only as a comparator. The row-naive p-value answers what would be concluded if selected rows were treated as independent. It is not the confirmatory test.

Fourth, use robust inference on the date series. The public implementation reports HAC standard errors, moving-block bootstrap, and stationary bootstrap evidence. The date-IID bootstrap is retained as a comparator.

Fifth, apply a fixed researcher-menu correction when several lookbacks, thresholds, or model variants were evaluated. A finite menu should be declared before the confirmatory run. In the replication code, Romano-Wolf stepdown is the primary fixed-menu adjustment.

Sixth, where row-level signals exist, run same-date placebo checks. Signal assignments can be permuted within date to test whether the selected rows carry information beyond the date-level common environment. If the permutation null is degenerate, it is reported as uninformative rather than as a pass or fail.

Seventh, apply a simple turnover-cost stress. The purpose is not to model execution at scale. The purpose is to check whether a fragile gross Sharpe claim survives a transparent cost drag.

Eighth, classify the result as robust, fragile, or failed rather than relying on a single favorable Sharpe statistic. A claim that passes only the row-naive test should be treated as false-precision evidence, not as a confirmatory portfolio result.

The same workflow can be summarized as portfolio-review pseudocode:

```
input: selected rows with date, asset, weight, and next-period return
for each date:
    aggregate selected rows into one portfolio return
    compute the date-level Sharpe ratio
    test the date-level Sharpe using HAC or a dependent bootstrap
    compute the row-naive Sharpe test only as a comparator
```

report the row/date uncertainty gap, placebo result, and cost stress
classify the backtest as robust, fragile, or failed

4 Worked examples with public data

The empirical section is a worked example, not a broad anomaly census. The goal is to show how a backtest reviewer would apply the checklist to public data. The row-level examples use Kenneth French portfolios because they permit date aggregation and same-date diagnostics. AQR factor series are included as useful time-series benchmarks, but they do not provide public constituent-level signal rows for same-date permutation or row-boundary diagnostics.

4.1 Case 1: canonical WML momentum survives

The first case is the canonical French WML momentum benchmark. This case is important because the diagnostic should not mechanically reject every positive result. The WML benchmark remains significant after date-level inference and the fixed-menu checks used here. It is therefore best read as a calibration check: the workflow can preserve a strong benchmark while still forcing the evidence to be reported at the portfolio-return boundary.

4.2 Case 2: a weaker panel claim fails

The second case is a French Size/BM skipped-momentum panel. This example is weaker. The date-HAC evidence is not significant, the same-date permutation does not support the claim, and the 5 bps turnover-cost stress produces a negative net Sharpe. A row-level or favorable gross Sharpe summary would be a poor basis for a portfolio claim here.

4.3 Case 3: AQR factor series are time-series benchmarks only

The AQR factor series are economically meaningful, but the public files used here are pre-aggregated factor returns. They are therefore evaluated as time-series benchmarks. They can be tested with HAC and bootstrap methods on the date-return series, but they cannot be used for the row-level same-date signal permutation or turnover diagnostics unless constituent-level signal and trading information are available.

The main empirical tables report the row-level panel candidates only. Pre-aggregated AQR factor series, benchmark-validation details, horizon sensitivity, and full robustness diagnostics are reported in the technical appendix.

The comparator table makes the standard-practice contrast explicit: row-naïve evidence is shown next to date-level HAC and bootstrap inference on the portfolio return

Table 1. Panel candidates and date-level evaluation status.

Panel candidate	Annualized SR	Date-HAC p+	Permutation p	Net SR (5 bps)	Status
French momentum deciles, rank-threshold panel	0.447	7.54e-05	N/A	0.447	Permutation uninformative; remaining applicable checks pass
French WML decile benchmark	0.497	5.53e-06	1.00e-04	0.497	Passes full evaluation
French 25 Size/BM, skipped-momentum panel	0.058	0.412	0.943	-0.050	Fails statistical and cost checks

Note: Annualized SR is the gross date-level Sharpe ratio. Date-HAC p+ is the one-sided HAC-delta positive-edge p-value computed on the date-level portfolio return series. Permutation p is the same-date signal-placebo p-value; N/A means the placebo was degenerate or otherwise uninformative for that threshold and is not counted as a full pass. Net SR (5 bps) is the Sharpe ratio after the reported 5 bps turnover-cost stress.

series.

Table 2. Row-naive and date-level inference comparisons for panel candidates.

Panel candidate	Row-naive p+	Date-IID p+	Date-HAC p+	Moving-block p+	Uncertainty infl.
French momentum deciles, rank-threshold panel	1.70e-04	2.00e-04	7.54e-05	2.00e-04	1.000
French WML decile benchmark	4.85e-04	2.00e-04	5.53e-06	2.00e-04	1.000
French 25 Size/BM, skipped-momentum panel	0.253	0.417	0.412	0.422	9.002

Note: p+ denotes a one-sided positive-Sharpe p-value. Row-naive p+ treats selected asset-date rows as independent. Date-IID, Date-HAC, and Moving-block p+ operate on the date-level portfolio return series. Uncertainty infl. is the diagnostic variance ratio between date-level long-run Sharpe uncertainty and row-pooled IID Sharpe uncertainty.

5 How large can the false-precision problem be?

The simulation is not intended to be a general theorem about all return processes. It is a stress test for a common backtesting workflow: treating selected asset-date rows as independent when portfolio performance is realized by date. The practical question is how often a row-naive reviewer would approve a strategy under a null design with no date-level edge.

The simulation varies same-date dependence and serial dependence while setting the population date-level Sharpe to zero in the null designs. The row-naive test treats selected entity-date rows as independent. The date-level procedures first aggregate rows into portfolio returns and then apply HAC, moving-block bootstrap, stationary bootstrap, and Romano-Wolf adjustments. All procedures are evaluated at nominal 5 percent size.

The null-size figure and table are the core simulation results: under dependent nulls, row-naive testing rejects about one third of the time at a nominal 5 percent level, while date-level HAC, block-bootstrap, stationary, and Romano-Wolf procedures stay much closer to size.

The result is economically large. In the dependent-null designs, row-naive testing rejects about 35 percent of the time at a nominal 5 percent level. Date-level dependence-aware procedures remain much closer to nominal size. The practical interpretation is direct: a backtest workflow that multiplies selected rows can approve strategies that do not have a true date-level edge in the data-generating process.

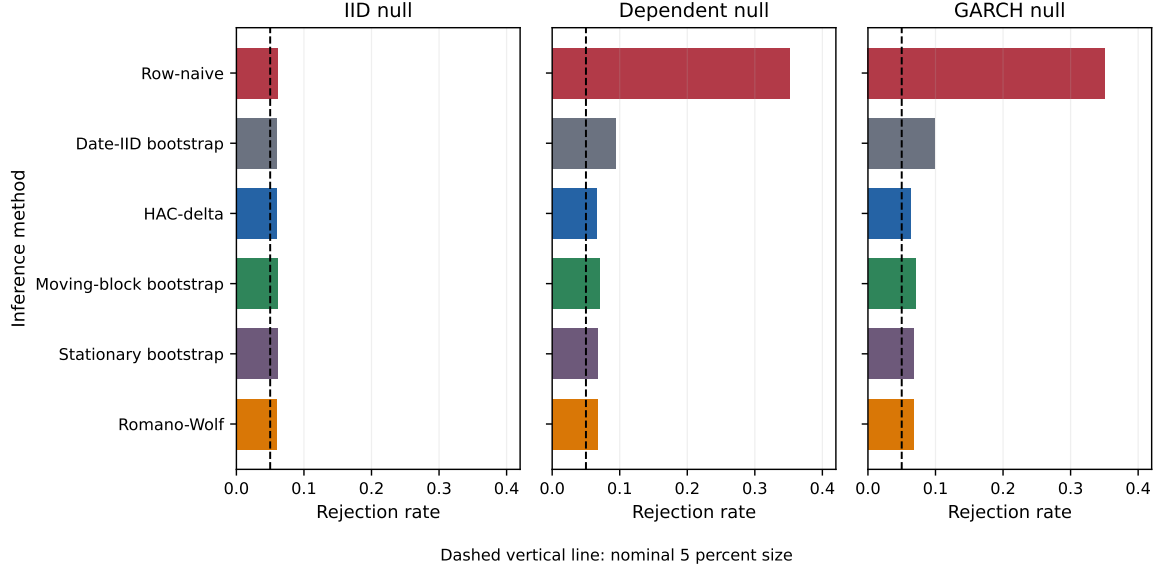


Figure 1. Rejection rates under null designs. The dashed line is the nominal 5 percent rejection rate. Row-naive rejection is near nominal under IID sampling but rises sharply under dependent nulls, while date-level dependence-aware procedures remain much closer to nominal size.

Table 3. Monte Carlo rejection rates under null designs with no true edge.

Null design	Row-naive	Date-IID bootstrap	HAC-delta	Moving-block bootstrap	Stationary bootstrap	Romano-Wolf
Dependent null	0.352	0.094	0.066	0.070	0.067	0.068
GARCH null	0.351	0.098	0.063	0.070	0.068	0.067
IID null	0.062	0.060	0.060	0.061	0.062	0.060

Note: Each entry is the fraction of 1000 Monte Carlo replications rejected at nominal 5 percent size. The main null designs use $T=1000$ dates and $N=50$ entities. Dependent null combines same-date and serial dependence; GARCH null adds conditional heteroskedasticity; IID null removes both. Row-naive treats selected asset-date rows as independent; the other procedures operate on the date-level portfolio return series.

6 Practical guidance and limitations

The diagnostic is a reporting check, not a guarantee that a strategy is tradable. A positive result should still be evaluated for economic interpretation, implementation constraints, liquidity, turnover, capacity, model decay, and out-of-sample performance. The diagnostic only asks whether Sharpe evidence was tested at the right sampling boundary.

For asset managers and backtest reviewers, the minimum reporting package for a panel-based Sharpe claim should include: the date-level portfolio return construction, date-level Sharpe inference, row-naïve results as a comparator, a researcher-menu correction, same-date placebo checks where row-level signals are available, a turnover-cost stress, and a clear classification of the result as robust, fragile, or failed.

The public evidence is illustrative. It uses public French and AQR datasets rather than a proprietary production signal library or a broad published-anomaly census. The simulation designs are stress tests for a specific reporting failure, not a complete model of financial returns. The HAC and bootstrap procedures also rely on standard stationarity, moment, and dependence conditions. Long memory, structural breaks, infinite-variance returns, and growing researcher menus require stronger assumptions or different methods.

7 Conclusion

Panel backtests can create false precision when selected rows are treated as independent performance observations even though the realized payoff is one portfolio return per date. The practical fix is straightforward: aggregate selected rows into date-level portfolio returns, test Sharpe evidence on that date series, and report how much precision the row-pooled analysis created. Public examples show that strong benchmark momentum can survive this workflow while weaker panel claims fail. Simulation stress tests show that the false-precision problem can be large enough to turn a nominal 5 percent test into a roughly 35 percent row-naïve rejection rate under dependent nulls. The diagnostic is not a new trading rule. It is a backtest-review workflow for reporting portfolio-performance evidence at the correct payoff boundary.

Data and code availability

For double-blind review, the public repository URL is omitted from the manuscript. An anonymized reproducibility package is supplied separately for refereeing; the public repository URL is provided on the title page and can be inserted into the manuscript after review if required. The reported examples use public Kenneth French and AQR data plus simulation code. Raw third-party data are not redistributed where source terms do not permit redistribution.

Acknowledgments, funding, and declarations

Acknowledgments are omitted from the review manuscript to preserve double-blind anonymity. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The author declares no competing interests. The study uses public aggregate financial data and simulations only; it does not involve human participants, animals, clinical data, or private personal data.

References

- [Andrews, 1991] Andrews, Donald W. K. 1991. Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica* 59(3): 817–858.
- [Carhart, 1997] Carhart, Mark M. 1997. On persistence in mutual fund performance. *Journal of Finance* 52(1): 57–82.
- [Fama and French, 1993] Fama, Eugene F., and Kenneth R. French. 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33(1): 3–56.
- [Harvey et al., 2016] Harvey, Campbell R., Yan Liu, and Heqing Zhu. 2016. ...and the cross-section of expected returns. *Review of Financial Studies* 29(1): 5–68.
- [Harvey and Liu, 2020a] Harvey, Campbell R., and Yan Liu. 2020a. False (and missed) discoveries in financial economics. *Journal of Finance* 75(5): 2503–2553. <https://doi.org/10.1111/jofi.12951>.
- [Harvey and Liu, 2020b] Harvey, Campbell R., and Yan Liu. 2020b. A census of the factor zoo. SSRN working paper. <https://doi.org/10.2139/ssrn.3341728>.
- [Ledoit and Wolf, 2008] Ledoit, Olivier, and Michael Wolf. 2008. Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance* 15(5): 850–859.
- [Lo, 2002] Lo, Andrew W. 2002. The statistics of Sharpe ratios. *Financial Analysts Journal* 58(4): 36–52.
- [Moulton, 1986] Moulton, Brent R. 1986. Random group effects and the precision of regression estimates. *Journal of Econometrics* 32(3): 385–397.
- [Newey and West, 1987] Newey, Whitney K., and Kenneth D. West. 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55(3): 703–708.

- [Petersen, 2009] Petersen, Mitchell A. 2009. Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies* 22(1): 435–480. <https://doi.org/10.1093/rfs/hhn053>.
- [Politis and Romano, 1994] Politis, Dimitris N., and Joseph P. Romano. 1994. The stationary bootstrap. *Journal of the American Statistical Association* 89(428): 1303–1313.
- [Romano and Wolf, 2005] Romano, Joseph P., and Michael Wolf. 2005. Stepwise multiple testing as formalized data snooping. *Econometrica* 73(4): 1237–1282.
- [White, 2000] White, Halbert. 2000. A reality check for data snooping. *Econometrica* 68(5): 1097–1126.